Rockchip Linux Memory Developer Guide

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Preface

Overview

This document introduces common commands related to memory in the Linux development process, including viewing total memory size, usage status, and reserved memory size.

Product Version

Chip Models	Kernel Version
ARM Series Chips	4.19

Target Audience

This document (this guide) is primarily intended for:

Technical Support Engineers

Software Development Engineers

Revision History

Version	Author	Date	Description
V1.0.0	Xu Jianqun	2021-01-06	Initial version

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1. Total Memory Information

1.1 meminfo

- cat /proc/meminfo

```
MemTotal:        1497696 kB
MemFree:          590524 kB
MemAvailable:     773252 kB
Buffers:             768 kB
Cached:           216892 kB
SwapCached:        14204 kB
Active:           260884 kB
Inactive:         176744 kB
Active(anon):     120620 kB
Inactive(anon):   106688 kB
Active(file):     140264 kB
Inactive(file):    70056 kB
Unevictable:        3224 kB
Mlocked:            3224 kB
SwapTotal:        748844 kB
SwapFree:         346156 kB
Dirty:               776 kB
Writeback:             0 kB
AnonPages:        220804 kB
Mapped:           194276 kB
Shmem:              4968 kB
KReclaimable:      43900 kB
Slab:             111060 kB
SReclaimable:      40648 kB
SUnreclaim:        70412 kB
KernelStack:       18496 kB
PageTables:        42096 kB
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:     1497692 kB
Committed_AS:   41751048 kB
VmallocTotal:   263061440 kB
VmallocUsed:       30368 kB
VmallocChunk:          0 kB
Percpu:             2528 kB
CmaTotal:         532480 kB
CmaFree:               0 kB
```

MemTotal = MemFree + Buffers + Cached

Swap: SwapTotal/SwapFree, enabled and allocated via the swap command in system scripts.

> The swap partition, often referred to as the exchange partition, is a special area on the hard drive. When physical memory is insufficient, the operating system moves temporarily unused data from memory to the swap partition, thereby freeing up sufficient memory space for currently running programs.

Slab: Current usage size of Slab, comprising reclaimable (SReclaimable) and unreclaimable (SUnreclaim) portions. Excessive growth may indicate a memory leak.

Vmalloc: VmallocTotal represents the total size, while VmallocUsed indicates the used size.

Cma: CmaTotal denotes the total size, and CmaFree represents the free size.

> CMA is borrowed by the system and thus considered used; consequently, CmaFree decreases.

1.2 sysrq-trigger

- echo m > /proc/sysrq-trigger

```
Mem-Info:
active_anon:43194 inactive_anon:26284 isolated_anon:0
 active_file:35286 inactive_file:17959 isolated_file:0
 unevictable:806 dirty:0 writeback:0 unstable:0
 slab_reclaimable:10164 slab_unreclaimable:17630
 mapped:48836 shmem:1280 pagetables:10550 bounce:0
 free:134653 free_pcp:717 free_cma:0
Node 0 active_anon:172776kB inactive_anon:105136kB active_file:141144kB
inactive_file:71836kB unevictable:3224kB isolated(anon):0kB isolated(file):0kB
mapped:195344kB dirty:0kB writeback:0kB shmeo
DMA32 free:538612kB min:4884kB low:24024kB high:25520kB active_anon:172776kB
inactive_anon:105136kB active_file:141144kB inactive_file:71836kB unevictable:3224kB
writepending:0kB present:2095104kB
lowmem_reserve[]: 0 0 0
DMA32: 13625*4kB (UEH) 12210*8kB (UEH) 8286*16kB (UMEH) 4151*32kB (UMEH) 1157*64kB
(UMEH) 223*128kB (UM) 46*256kB (U) 9*512kB (U) 2*1024kB (U) 0*2048kB 0*4096kB =
538612kB
58687 total pagecache pages
3564 pages in swap cache
Swap cache stats: add 155134, delete 151579, find 11930/52536
Free swap  = 346412kB
Total swap = 748844kB
523776 pages RAM
0 pages HighMem/MovableOnly
149352 pages reserved
133120 pages cma reserved
```

2. Kernel Memory Information

2.1 slab

Kernel memory allocation debugging interface; two macro definitions must be enabled during compilation.

```
CONFIG_SLUB_SYSFS=y
CONFIG_SLUB_DEBUG=y
```

Command to calculate the total slab size

```
echo `cat /proc/slabinfo |awk 'BEGIN{sum=0;}{sum=sum+$3*$4;}END{print sum/1024/1024}'` MB
```

2.1.1 kernfs_node_cache

The following command shows that kernfs uses 2.5M, where 655 pages = 2.55859375 MBytes.

```
[root@RV1126_RV1109:/]# cat /proc/slabinfo |grep kernfs_node_cache
kernfs_node_cache  18319  18340    144   28    1 : tunables    0    0    0 : slabdata
655    655      0
```

The code is located at `fs/kernfs/mount.c`.

```
void __init kernfs_init(void)
{

    /*
     * the slab is freed in RCU context, so kernfs_find_and_get_node_by_ino
     * can access the slab lock free. This could introduce stale nodes,
     * please see how kernfs_find_and_get_node_by_ino filters out stale
     * nodes.
     */
    kernfs_node_cache = kmem_cache_create("kernfs_node_cache",
                          sizeof(struct kernfs_node),
                          0,
                          SLAB_PANIC | SLAB_TYPESAFE_BY_RCU,
                          NULL);
}
```

2.1.2 inode_cache

The following command shows that the inode cache uses 3.2M, where 816 pages = 3.1875 MBytes.

```
[root@RV1126_RV1109:/]# cat /proc/slabinfo |grep inode_cache
inode_cache         6282   6340    408   20    2 : tunables    0    0    0 : slabdata
317    317      0
```

The code is located in `fs/inode.c`.

```
void __init inode_init(void)
{
    /* inode slab cache */
    inode_cachep = kmem_cache_create("inode_cache",
                    sizeof(struct inode),
                    0,
                    (SLAB_RECLAIM_ACCOUNT|SLAB_PANIC|
                    SLAB_MEM_SPREAD|SLAB_ACCOUNT),
                    init_once);
}
```

2.2 memblock

System reserved memory debugging interface. To enable, add `memblock=debug` to bootargs during compilation.

```
chosen {
        bootargs = "earlycon=uart8250,mmio32,0xff570000 console=ttyFIQ0 memblock=debug";
};
```

```
[    0.000000] MEMBLOCK configuration:
[    0.000000]  memory size = 0x3fdb8000 reserved size = 0x11706eb8
[    0.000000]  memory.cnt  = 0x2
[    0.000000]  memory[0x0]     [0x00000000-0x083fffff], 0x08400000 bytes flags: 0x0
[    0.000000]  memory[0x1]     [0x08648000-0x3fffffff], 0x379b8000 bytes flags: 0x0
[    0.000000]  reserved.cnt  = 0x7
[    0.000000]  reserved[0x0]   [0x00004000-0x00007fff], 0x00004000 bytes flags: 0x0
[    0.000000]  reserved[0x1]   [0x00100000-0x00e32b2f], 0x00d32b30 bytes flags: 0x0
[    0.000000]  reserved[0x2]   [0x08000000-0x080fffff], 0x00100000 bytes flags: 0x0
[    0.000000]  reserved[0x3]   [0x08300000-0x08318fff], 0x00019000 bytes flags: 0x0
[    0.000000]  reserved[0x4]   [0x2dc00000-0x3dbfffff], 0x10000000 bytes flags: 0x0 CMA
[    0.000000]  reserved[0x5]   [0x3df00000-0x3dfb7387], 0x000b7388 bytes flags: 0x0
[    0.000000]  reserved[0x6]   [0x3f800000-0x3fffffff], 0x00800000 bytes flags: 0x0 CMA
```

The largest reserved regions include initrd (used for loading the root filesystem) and node memmap (used for managing page memory).

The initrd reservation is 13 MBytes.

```
void __init arm_memblock_init(const struct machine_desc *mdesc)
{
        arm_initrd_init();
}
```

The node memmap reserves 8 MBytes.

```
[    0.000000] On node 0 totalpages: 262006
[    0.000000] memblock_reserve: [0x3ef75000-0x3f774fff]
memblock_virt_alloc_internal+0x108/0x1a4
[    0.000000] alloc_node_mem_map: node 0, pgdat b0d4bd00, node_mem_map eef75000
[    0.000000]   Normal zone: 2048 pages used for memmap
[    0.000000]   Normal zone: 0 pages reserved
[    0.000000]   Normal zone: 262006 pages, LIFO batch:63
```

Device memory status printed at boot:

```
[    0.000000] Virtual kernel memory layout:
[    0.000000]     vector  : 0xffff0000 - 0xffff1000   (   4 kB)
[    0.000000]     fixmap  : 0xffc00000 - 0xfff00000   (3072 kB)
[    0.000000]     vmalloc : 0xf0800000 - 0xff800000   ( 240 MB)
[    0.000000]     lowmem  : 0xb0000000 - 0xf0000000   (1024 MB)
[    0.000000]     pkmap   : 0xafe00000 - 0xb0000000   (   2 MB)
[    0.000000]     modules : 0xaf000000 - 0xafe00000   (  14 MB)
[    0.000000]       .text : 0x(ptrval) - 0x(ptrval)   (9184 kB)
[    0.000000]       .init : 0x(ptrval) - 0x(ptrval)   (1024 kB)
[    0.000000]       .data : 0x(ptrval) - 0x(ptrval)   ( 332 kB)
[    0.000000]        .bss : 0x(ptrval) - 0x(ptrval)   ( 896 kB)
```

```
[    0.000000] Memory: 752320K/1048024K available (8192K kernel code, 331K rwdata, 1900K
rodata, 1024K init, 895K bss, 25368K reserved, 270336K cma-reserved, 0K highmem)
```

752320K: Free page available memory, `nr_free_pages() << (PAGE_SHIFT - 10)`

1048024K: Total visible physical memory, `physpages << (PAGE_SHIFT - 10)`

kernel code: Kernel code, `codesize = _etext - _stext`

rwdata: Read-write data segment, `datasize = _edata - _sdata`

rodata: Read-only data segment, `rosize = __end_rodata - __start_rodata`

init: Includes `init_data_size = __init_end - __init_begin` and `init_code_size = _einittext - _sinittext`

bss: Memory region used to store uninitialized global variables in the program, `bss_size = __bss_stop - __bss_start`

reserved: memblock reserved memory, `physpages - totalram_pages - totalcma_pages`

cma-reserved: CMA reserved memory, `totalcma_pages`

```
[    0.000000] Reserved memory: created CMA memory pool at 0x3f800000, size 8 MiB
[    0.000000] OF: reserved mem: initialized node linux,cma, compatible id shared-dma-pool
[    0.000000] Reserved memory: created CMA memory pool at 0x2dc00000, size 256 MiB
[    0.000000] OF: reserved mem: initialized node isp, compatible id shared-dma-pool
```

2.2.1 memory

The kernel supports multiple memory blocks. On Rockchip platforms, certain memory regions are reserved for specific purposes before entering the kernel. For example, OPTEE/ATF code must be placed in the 132MB–135MB range. As shown below, a memory block of size 0x248000 is reserved for OPTEE code storage.

```
[root@RV1126_RV1109:/]# cat /sys/kernel/debug/memblock/memory
   0: 0x00000000.0x083fffff
   1: 0x08648000.0x3fffffff
```

2.2.2 reserved

```
[root@RV1126_RV1109:/]# cat /sys/kernel/debug/memblock/reserved
   0: 0x00004000.0x00007fff
   1: 0x00100000.0x00e32b2f
   2: 0x08000000.0x080fffff
   3: 0x08300000.0x08318fff
   4: 0x083ff000.0x083fffff
   5: 0x2dc00000.0x3dbfffff
   6: 0x3df00000.0x3dfb7387
   7: 0x3ee41000.0x3ef3cfff
   8: 0x3ef3f480.0x3ef3f4f7
   9: 0x3ef3f500.0x3ef3f803
  10: 0x3ef3f834.0x3f7fefff
  11: 0x3f7ff040.0x3f7ff384
  12: 0x3f7ff3c0.0x3f7ff3fb
  13: 0x3f7ff400.0x3f7ff583
  14: 0x3f7ff5c0.0x3f7ff784
  15: 0x3f7ff7c0.0x3f7ff837
  16: 0x3f7ff840.0x3f7ff84f
  17: 0x3f7ff880.0x3f7ff88f
  18: 0x3f7ff8c0.0x3f7ff8c3
  19: 0x3f7ff900.0x3f7ff903
  20: 0x3f7ff940.0x3f7ffa4b
  21: 0x3f7ffa80.0x3f7ffb8b
  22: 0x3f7ffbc0.0x3f7ffccb
  23: 0x3f7ffcf0.0x3f7ffd08
  24: 0x3f7ffd0c.0x3f7ffd24
  25: 0x3f7ffd28.0x3f7ffd72
  26: 0x3f7ffd74.0x3f7ffd8e
  27: 0x3f7ffd90.0x3f7ffdaa
  28: 0x3f7ffdac.0x3f7ffdc6
  29: 0x3f7ffdc8.0x3f7ffde2
  30: 0x3f7ffde4.0x3f7ffdfe
  31: 0x3f7ffe00.0x3f7ffedf
  32: 0x3f7ffee8.0x3f7fff9f
  33: 0x3f7fffb0.0x3fffffff
```

2.3 zoneinfo

- cat /proc/zoneinfo

```
Node 0, zone    DMA32
  per-node stats
      nr_inactive_anon 28250
      nr_active_anon 34245
      nr_inactive_file 29724
      nr_active_file 33943
      nr_unevictable 806
      nr_slab_reclaimable 10088
      nr_slab_unreclaimable 17495
      nr_isolated_anon 0
      nr_isolated_file 0
      workingset_refault 258089
      workingset_activate 54883
      workingset_restore 35120
      workingset_nodereclaim 4574
      nr_anon_pages 59384
      nr_mapped    65408
      nr_file_pages 70824
      nr_dirty     31
      nr_writeback 0
      nr_writeback_temp 0
      nr_shmem     1581
      nr_shmem_hugepages 0
      nr_shmem_pmdmapped 0
      nr_anon_transparent_hugepages 0
      nr_unstable  0
      nr_vmscan_write 148946
      nr_vmscan_immediate_reclaim 2540
      nr_dirtied   798484
      nr_written   938957
      nr_kernel_misc_reclaimable 0
      nr_unreclaimable_pages 0
      nr_ion_heap  0
      nr_ion_heap_pool 0
      nr_gpu_heap  0
  pages free     133183
        min      1221
        low      6006
        high     6380
        spanned  523776
        present  523776
        managed  374424
        protection: (0, 0, 0)
      nr_free_pages 133183
      nr_zone_inactive_anon 28250
      nr_zone_active_anon 34245
      nr_zone_inactive_file 29724
      nr_zone_active_file 33943
      nr_zone_unevictable 806
      nr_zone_write_pending 31
      nr_mlock     806
```

```
      nr_page_table_pages 10425
      nr_kernel_stack 18256
      nr_bounce    0
      nr_zspages   26061
      nr_free_cma  0
  pagesets
    cpu: 0
              count: 342
              high:  378
              batch: 63
  vm stats threshold: 30
    cpu: 1
              count: 301
              high:  378
              batch: 63
  vm stats threshold: 30
    cpu: 2
              count: 32
              high:  378
              batch: 63
  vm stats threshold: 30
    cpu: 3
              count: 296
              high:  378
              batch: 63
  vm stats threshold: 30
  node_unreclaimable:  0
  start_pfn:           512
Node 0, zone   Normal
  pages free     0
        min      0
        low      0
        high     0
        spanned  0
        present  0
        managed  0
        protection: (0, 0, 0)
Node 0, zone  Movable
  pages free     0
        min      0
        low      0
        high     0
        spanned  0
        present  0
        managed  0
        protection: (0, 0, 0)
```

3. Free Memory

3.1 free

- total: Total memory managed by the kernel page allocator.
- used: Memory currently in use.
- free: Idle memory.
- shared: Shared memory, such as anonymous pages.
- buffers: Buffered memory.
- cached: Cached memory.

> Difference between buffers and cached. Cached accelerates data reads, while buffers accelerate data writes. Cached refers to storing read data in memory so that subsequent reads can be served directly from memory instead of the disk, accelerating the read process. Buffers refer to temporarily storing scattered write operations in memory and flushing them to the disk in batches once a certain threshold is reached, reducing disk fragmentation and repeated disk seeks, thereby accelerating the write process.

free

```
                 total        used        free      shared     buffers
  Mem:       1533640704  1308901376   224739328     6422528      921600
-/+ buffers/cache:     1307979776   225660928
  Swap:       766816256   401342464   365473792
```

free -m

```
                 total        used        free      shared     buffers
  Mem:             1462        1249         213           6           0
-/+ buffers/cache:           1248         214
  Swap:             731         382         348
```

free -h

```
                 total        used        free      shared     buffers
  Mem:             1.4G        1.2G        212M        6.1M        900K
  -/+ buffers/cache:           1.2G        213M
  Swap:            731M        383M        349M
```

4. Clearing Cache

4.1 drop_caches

- echo x > /proc/sys/vm/drop_caches

Clear the pagecache

```
echo 1 > /proc/sys/vm/drop_caches
```

Clear dentries and inodes

```
echo 2 > /proc/sys/vm/drop_caches
```

Clear all caches (pagecache, dentries, and inodes)

```
echo 3 > /proc/sys/vm/drop_caches
```

5. Virtual Memory Usage

5.1 vmallocinfo

- cat /proc/vmallocinfo

```
0x0000000000000000-0x0000000000000000    8192 bpf_jit_binary_alloc+0x70/0x110 pages=1
vmalloc
...
0x0000000000000000-0x0000000000000000   20480 start_kernel+0x330/0x4f0 pages=4 vmalloc
0x0000000000000000-0x0000000000000000    8192 of_iomap+0x4c/0xb8 phys=0x00000000fdd00000
ioremap
0x0000000000000000-0x0000000000000000   20480 start_kernel+0x330/0x4f0 pages=4 vmalloc
0x0000000000000000-0x0000000000000000    8192 of_iomap+0x4c/0xb8 phys=0x00000000fdd20000
ioremap
0x0000000000000000-0x0000000000000000   20480 start_kernel+0x330/0x4f0 pages=4 vmalloc
```

5.2 vmstat

- cat /proc/vmstat

6. Memory Pressure

6.1 pressure

- cat /proc/pressure/memory
- **VSS** - Virtual Set Size: Virtual memory consumed (includes memory occupied by shared libraries)
- **RSS** - Resident Set Size: Physical memory actually used (includes memory occupied by shared libraries)

- **PSS** - Proportional Set Size: Physical memory actually used (shared library memory allocated proportionally)
- **USS** - Unique Set Size: Physical memory exclusively occupied by the process (excludes memory occupied by shared libraries)

Generally, memory usage follows this pattern: VSS >= RSS >= PSS >= USS